

used on a full-frame or 35mm film camera, has the same angle of view as that lens does on that digital camera. For example, a 100mm lens on a 1.5x factor camera shows the same area of view that a 150mm lens would show on a 35mm film or full-frame camera.

A commonly asked question is “what camera sensor resolution is suitable for me?” And a quick answer to this is it all depends on your purpose or use of the images generated. For posting online or for newspaper reproductions, you can pick any current dSLR for sufficient resolution. When you’ll need to crop for details or need larger print sizes or clearer pictures say, for magazine printing, somewhere around 8 megapixels will be a good starting point. dSLRs above 10 megapixels will do the trick for fine-art landscape shutterbugs and others seeking maximum details. This is a generic answer for generic use but if you have a specific target size in sight, there’s a commonly followed method you can apply with simple math. Let’s say you want to take an 8 x 10 inch print on an inkjet printer. Firstly, know the required output resolution; let’s assume this to be 300 pixels per inch or greater. Now, multiply the needed output resolution by the linear dimensions of your final print. In this example that equals 8 inches x 300 PPI = 2,400 pixels needed for the vertical dimension and 10 inches x 300 PPI = 3,000 pixels for the horizontal. Now multiply the horizontal and vertical, here $2,400 \times 3,000 = 7,200,000$ or 7.2 megapixels. It’s advisable to take an upper margin here so that you can crop a picture later without compromising on resolution. Say we estimate 20 percent of a photo is generally cropped, so we add that to our required value, giving us $7.2 \times 1.2 = 8.64$ megapixels. Thus approximately a good minimum resolution will be something above 8.5 megapixels.

While caring for sensors, the most common problem is dust. Whenever you change the lens in your dSLR, make sure you don’t get any dust on the sensor as this can result in bad pixellation or too much noise. At best, you can cover up for it by re-touching the images during post processing; at worst, you can seriously mess up your camera’s internals. Reasonable care while swapping lenses and not too rough handling of your camera is sufficient to prevent dust from being an issue for you, unless of course you’re planning to shoot in sandy or particulate heavy terrain. For everyday use don’t worry, because a combination of anti-dust technologies are present today in almost all SLRs, like vibrating the sensor on startup to dislodge any settled particles.